

Endoresection Versus Iodine-125 Plaque Brachytherapy for the Treatment of Choroidal Melanoma

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- **PURPOSE:** To compare the effectiveness of pars plana vitrectomy endoresection with iodine-125 brachytherapy in the treatment of choroidal melanoma.
- **DESIGN:** A nested case-control study (1:2) from a surgical cohort.
- **METHODS:** The study comprised 81 choroidal melanoma patients treated with either endoresection or iodine-125 brachytherapy. Twenty-seven patients who had undergone endoresection were matched according to tumor height and postequatorial tumor location with 54 cases treated with iodine-125 brachytherapy. Metastatic disease, overall survival, local tumor recurrence, visual acuity, and secondary enucleation rates were analyzed and compared between groups.
- **RESULTS:** Metastatic spread was observed in 11 patients in the iodine-125 brachytherapy group vs only 1 patient in the endoresection group (20.4% and 3.7%, respectively, $P = .053$). Fourteen patients died during follow-up: 11 in the brachytherapy arm vs 3 in the endoresection arm (20.4% and 11.1%, respectively, $P = .238$). For the iodine-125 brachytherapy and endoresection groups, respectively, the 5-year Kaplan-Meier estimates were as follows: overall survival, 81.5% vs 89.2% (log-rank test, $P = .429$); relapse-free survival, 96.6% vs 92.4% ($P = .2$); visual acuity retention equal or superior to 20/200, 66.4% vs 59.9% ($P = .083$), and eye retention, 85.7% vs 87.8% ($P = .942$).
- **CONCLUSIONS:** Endoresection for choroidal melanoma is an effective treatment modality in selected cases of posterior choroidal melanomas, with outcomes similar to those obtained with iodine-125 brachytherapy. (Am J Ophthalmol 2013;156:334–342. © 2013 by Elsevier Inc. All rights reserved.)

THE TREATMENT OF POSTERIOR CHOROIDAL MELANOMA remains challenging and the optimal approach is still unclear. Most treatments are eye-preserving techniques such as transpupillary thermotherapy, plaque brachytherapy, proton-beam radiotherapy, trans-scleral tumor resection, and endoresection. The most widely used conservative treatment for uveal melanoma is iodine-125 brachytherapy. The local tumor recurrence rate after iodine-125 brachytherapy can be as low as 10%,¹ with eye retention in approximately 80% of patients, even in the case of large tumors.²

Peyman and associates^{3,4} were among the first authors to describe an alternative conservative approach in which internal resection is performed via the pars plana using vitrectomy instruments. Later, Damato and associates⁵ treated 52 uveal melanoma patients using this same internal approach, which they termed “endoresection.” García-Arumi and associates^{6,7} reported the long-term outcomes of this technique in a group of patients treated for high posterior choroidal melanoma.

Pars plana vitrectomy endoresection has been described as a primary procedure for tumors in patients who are not expected to have good outcomes after more conventional forms of treatment or as a secondary procedure used after toxic tumor syndrome.⁵ Some authors advocate the use of neoadjuvant proton beam or stereotactic radiotherapy before endoresection.^{8,9} Whether used alone or with adjuvant radiotherapy, endoresection appears to be a reasonable approach to a rapidly growing melanoma located in the posterior pole, especially when the tumor exhibits pronounced height and a small base diameter. Some authors consider endoresection to be preferable in these cases because of the large tumor size, the added morbidity associated with the proximity of the tumor to the optic disc/fovea, and the difficulty of treating these tumors with radiotherapy modalities.⁷ However, endoresection remains a highly controversial technique because of the theoretical risk of tumor dissemination when using this approach, although such concerns have been voiced for radiotherapy and enucleation as well.¹⁰

The aim of this study was to compare endoresection to iodine-125 brachytherapy in patients with uveal melanomas limited to the equatorial area. We report the comparative results in terms of metastatic disease, overall patient survival, local tumor recurrence, visual function, and eye retention.

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TABLE 1. Endoresection Versus Iodine-125 Plaque Brachytherapy in the Treatment of Choroidal Melanoma: Clinical and Demographic Characteristics of Enrolled Patients (Quantitative Variables)

	Brachytherapy (n = 54)	Endoresection (n = 27)	P Value ^b
	Median (IQR)	Median (IQR)	
Age (y) ^a	55.99 (14.66)	64.05 (15.05)	.027
Visual acuity (Snellen chart)	0.5 (0.7)	0.2 (0.5)	.084
Visual acuity (logMAR)	0.301 (0.903)	0.699 (2.78)	.084
Height (mm) ^a	6.95 (1.8)	7.69 (2.27)	.117
Tumor horizontal base (mm)	12.0 (3.0)	10.0 (3.0)	<.001
Tumor vertical base (mm)	12.0 (2.5)	10.0 (4.0)	.020
Tumor maximum diameter base (mm)	12.5 (2.0)	11.0 (2.9)	<.001
Fovea distance (mm) ^a	5.22 (3.52)	3.33 (2.93)	.020
Optic nerve distance (mm) ^a	5.49 (3.31)	2.80 (2.99)	.001

IQR = interquartile range.

^aNormal distribution, mean (SD).

^bP values were obtained by Mann-Whitney U test or Student t test as needed.

PATIENTS AND METHODS

THIS WAS A NESTED CASE-CONTROL STUDY (1:2) BASED ON a historical surgical cohort. From September 1995 through May 2010, 416 patients diagnosed with uveal melanoma were treated at the Ocular Oncology Unit of the Ophthalmology Department of Bellvitge University Hospital and at the Brachytherapy Department of the Catalan Oncology Institute. We selected 27 consecutive patients who had undergone endoresection and matched them according to tumor height and postequatorial tumor location with 54 patients who had been treated with iodine-125 brachytherapy. Patient confidentiality was protected by national data confidentiality laws. This manuscript has been revised for publication by the Clinical Research Ethics Committee of Bellvitge University Hospital (IRB00005523). Written informed consent was obtained from all subjects in accordance with our institutional guidelines. This study adhered to the tenets of the Declaration of Helsinki.

All patients included in the study presented uveal melanomas located close to the optic nerve and/or fovea. In all cases, the anterior margins did not extend beyond the equator of the eye. None of the patients had scleral involvement or evidence of distant metastasis at the time of study inclusion. The tumor characteristics for both groups are presented in Tables 1 and 2.

• **TREATMENT AND FOLLOW-UP:** Patients were fully informed of all relevant aspects of the procedures,

TABLE 2. Endoresection Versus Iodine-125 Plaque Brachytherapy in the Treatment of Choroidal Melanoma: Clinical and Demographic Characteristics of Enrolled Patients (Qualitative Variables)

	Brachytherapy (n = 54)		Endoresection (n = 27)		P Value ^c
	n	%	n	%	
Sex					
Men	27	50.0	13	48.1	.875
Women	27	50.0	14	51.9	
Eye					
Right	29	53.7	14	51.9	.672
Left	22	40.7	13	48.1	
Tumor size ^a					
Medium	50	92.6	22	81.5	.152
Large	4	7.4	5	18.5	
Tumor size ^b					
T1	1	1.9	1	3.7	.149
T2	26	48.1	17	63.0	
T3	27	50.0	9	33.3	
Retinal detachment					
No	24	44.4	6	22.2	.051
Yes	30	55.6	21	77.8	
Visual acuity					
20/200 or better	42	77.8	18	66.7	.058
Inferior to 20/200	12	22.2	9	33.3	
Bruch membrane rupture					
No	33	61.1	3	11.1	<.001
Yes	21	38.9	24	88.9	

^aAccording to COMS classification.

^bAccording to the 7th edition of AJCC staging.

^cP values were obtained by χ^2 or Fisher exact test as needed.

including the controversy surrounding endoresection. The preoperative examination included a best-corrected Snellen visual acuity test, a complete ophthalmoscopic examination, and A- and B-scans to measure the maximum tumor thickness and diameter. Tumor size was classified according to the criteria established by the Collaborative Ocular Melanoma Study (COMS) and according to the tumor size categories (T) published in the 7th edition of the American Joint Committee of Cancer (AJCC) staging system. All patients underwent screening for metastatic disease, which included serum biochemistry with liver function determinations, ultrasound evaluation of the liver, chest radiography, and evaluation by an oncologist.

Brachytherapy was administered using iodine-125 COMS-type plaques. Trans-scleral illumination was used to assess tumor margins with the patient under local anesthesia. In all cases, a minimum safety margin of 2 mm was employed and the same surgeon (J.M.C.) performed all procedures in the Brachytherapy Department. A total dose of 85 Gy was delivered to the apex of the tumor using a dose rate of 50-120 cGy/hour. Initial follow-up was

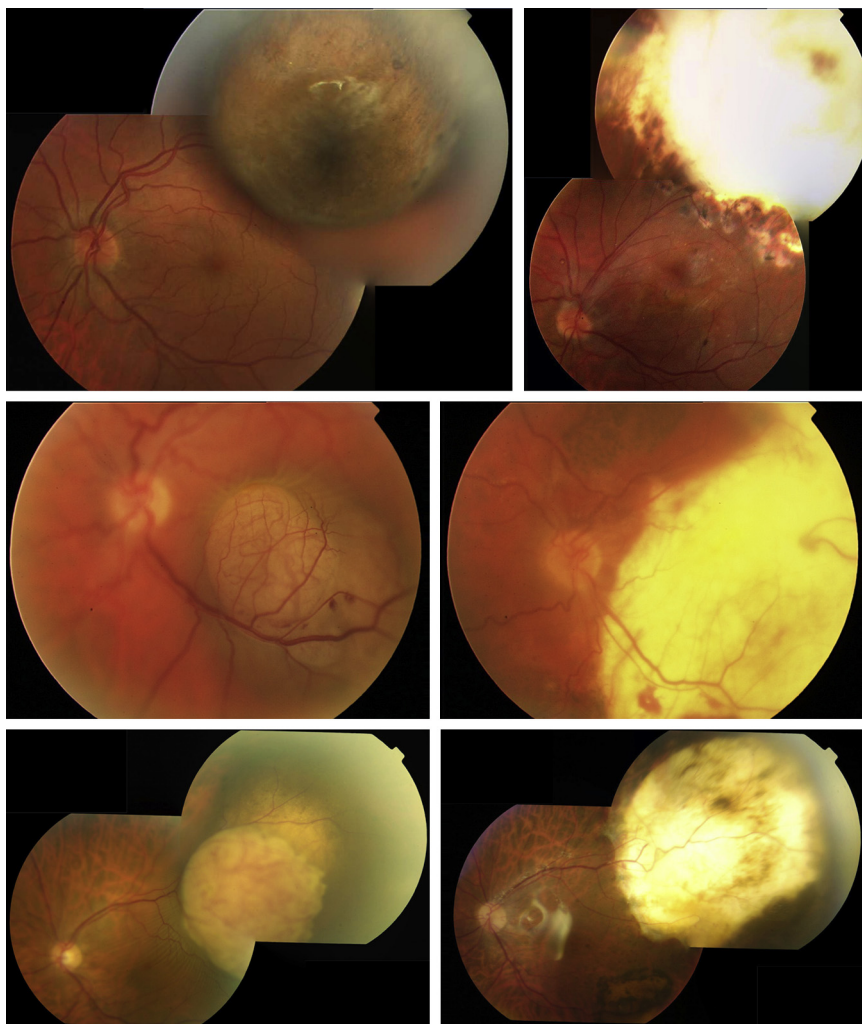


FIGURE 1. Illustration of 3 patients with uveal melanomas who underwent endoresection. (Top row, left) Patient with a superior temporal choroidal melanoma in the left eye (height, 8.4 mm; maximum basal diameter, 9 mm) with a visual acuity of 20/40 at the time of diagnosis. (Top row, right) The same patient after surgery, with no evidence of regrowth during follow-up (84 months) and a visual acuity of 20/200. (Middle row, left) Patient with an inferior macular choroidal melanoma with foveal detachment in the left eye (height, 5.2 mm; maximum basal diameter, 10 mm) with a visual acuity inferior of 20/200 at the time of diagnosis. (Middle row, right) The same patient after surgery, with no evidence of regrowth during follow-up (97 months) and a visual acuity worse than 20/200. (Bottom row, left) Patient with a superior temporal choroidal melanoma in the left eye with macular detachment (height, 7.5 mm; maximum basal diameter, 9 mm) with a visual acuity worse than 20/200 at the time of diagnosis. (Bottom row, right) The same patient after surgery, with no evidence of regrowth during follow-up (36 months) and a visual acuity of 20/200.

performed approximately 15 days after the intervention and at 3 months; thereafter, follow-up assessment, including screening for metastasis, was performed every 6 months during the first 5 years and annually thereafter.

In the case of endoresection, the surgical technique varied according to the degree of tumor invasion. In cases where the tumor had not invaded the retina, endoresection consisted of 20- to 23-gauge vitrectomy using a panoramic viewing system, followed by posterior hyaloid dissection, large retinotomy anterior to the tumor, and laser endophotocoagulation with 600-800 mW directed 2 mm beyond the tumor margins using the continuous mode of a 532-nm

laser. The melanoma lesion was removed with the vitrectomy probe using a bimanual technique to separate and protect the retina. Intraoperative choroidal bleeding was controlled by temporarily raising the intraocular pressure (IOP) to 60-70 mm. In all cases, patient blood pressure remained within normal limits during the procedure. Tumor excision was performed from the tumor apex to the scleral bed. Cellular remnants at the scleral bed were photocoagulated with high doses (800-1000 mW) of continuous-mode laser endophotocoagulation at 532 nm. The retina was reattached with perfluorocarbon liquid. Laser retinopexy endophotocoagulation was performed at

TABLE 3. Endoresection Versus Iodine-125 Plaque Brachytherapy in the Treatment of Choroidal Melanoma: Treatment Outcome and Follow-up

	Brachytherapy (n = 54)		Endoresection (n = 27)		P Value ^a
	n	%	n	%	
Metastatic disease					
No	43	79.6	26	96.3	.053
Yes	11	20.4	1	3.7	
Observed death					
No	43	79.6	24	88.9	.238
Yes	11	20.4	3	11.1	
Specific death					
No	44	81.5	26	96.3	.089
Yes	10	18.5	1	3.7	
Local recurrence					
No	53	98.2	25	92.6	.256
Yes	1	1.8	2	7.4	
Enucleation					
No	47	87.0	24	88.9	≥.999
Yes	7	13.0	3	11.1	
Enucleation cause					
Retinal detachment	1	1.9	0	0	.057
Neovascular glaucoma	5	9.3	0	0	
Phthisis bulbi	0	0	1	3.7	
Local recurrence	1	1.9	2	7.4	
Enucleation cause					
Complication	6	11.1	1	3.7	.183
Local recurrence	1	1.9	2	7.4	
Final VA					
Inferior of 20/200	35	64.8	21	77.8	.234
20/200 or better	19	35.2	6	22.2	
Final VA (logMAR)					
Median (IQR)	2.01 (1.17)		2.56 (0.83)		.234
Follow-up (mo)					
Median (IQR)	70.52 (37.15)		59.37 (33.64)		.193

IQR = interquartile range; VA = visual acuity.

^aP values were obtained by χ^2 or Fisher exact test as needed for qualitative variables, and by Mann-Whitney *U* test for quantitative variables.

the margins of the retinotomy, and this procedure was followed by fluid-air exchange and silicone oil-air exchange (1300-5000 centistokes). In cases where the tumor had invaded the retina, continuous-mode laser endophotocoagulation at 532 nm was employed through the retina and around the tumor margins, and then both the tumor and the retina were removed without peripheral retinotomy. In all cases, the lens of the eye was removed before vitrectomy by phacoemulsification, and a 3-piece foldable acrylic intraocular lens was inserted. Patients were instructed to maintain a face-down position for 1 week after surgery. Postoperative follow-up was performed at 1 day, 1 week, 1 month, 3 months, and 6 months. Thereafter, follow-up, including screening for metastasis, was performed at 6-month intervals (Figure 1).

• **STATISTICS:** Descriptive analysis was performed, and measures of the central value (mean and median) and dispersion (standard deviation, interquartile range) were determined depending on whether there was a normal distribution (Kolmogorov-Smirnov test). For quantitative variables, the Student *t* test or the Mann-Whitney *U* test was used, as appropriate, to compare baseline characteristics resulting from tumor endoresection or iodine-125 brachytherapy. To evaluate differences in proportions, the χ^2 test or Fisher exact test was employed as needed.

A Kaplan-Meier analysis was performed to determine predictive variables of overall survival. Those variables that showed a statistically significant association with survival by the log-rank test ($P < .05$) were subsequently analyzed using a Cox regression model. Effect estimates were expressed as hazard ratios (HR) with 95% confidence intervals (95% CI). All statistical tests were 2-tailed, and *P* values of $\leq .05$ were deemed significant. All analyses were performed using SPSS 12.0 software (SPSS, Inc, Chicago, Illinois, USA).

RESULTS

A TOTAL OF 81 PATIENTS (54 TREATED BY IODINE-125 brachytherapy and 27 with endoresection) were evaluated. Patients in the iodine-125 brachytherapy and endoresection groups presented with comparable tumor heights and locations (posterior) and were comparable in terms of sex, eye affected, follow-up, and tumor size according to the COMS criteria and the T categories of the 7th edition of the AJCC staging system. Most tumors in both groups were classified as medium-sized (92.6% of the iodine-125 brachytherapy group and 81.5% of the endoresection group; $P = .152$) and as T2 or T3 (98% of the iodine-125 brachytherapy group and 96% of the endoresection group; $P = .149$). The mean follow-up period was 59.37 ± 33.64 months (range, 12-138 months) for the endoresection group and 70.52 ± 37.15 months (range, 12-148 months) for the iodine-125 brachytherapy group (Tables 1-3).

Despite the similarity between the 2 groups, several differences were observed. Patients in the iodine-125 brachytherapy group were younger (55.99 ± 14.66 years vs 64.05 ± 15.05 years; $P = .027$) and had a larger mean maximum basal tumor diameter (12.5 ± 2 mm vs 11 ± 2.9 mm; $P < .001$). In addition, the tumors of patients in the endoresection group were located closer to the optic disc (distance of 2.80 ± 2.99 mm vs 5.49 ± 3.31 mm; $P = .001$) and the fovea (distance of 3.33 ± 2.93 mm vs 5.22 ± 3.52 mm; $P = .020$) when compared to tumors in the brachytherapy group. Most patients in the endoresection group presented with a Bruch membrane rupture (89% of the endoresection group and 39% of the brachytherapy group; $P < .001$) (Tables 1 and 2).

Patients treated by iodine-125 brachytherapy received a median dose of 85 ± 7.3 Gy (range, 69-103 Gy) delivered

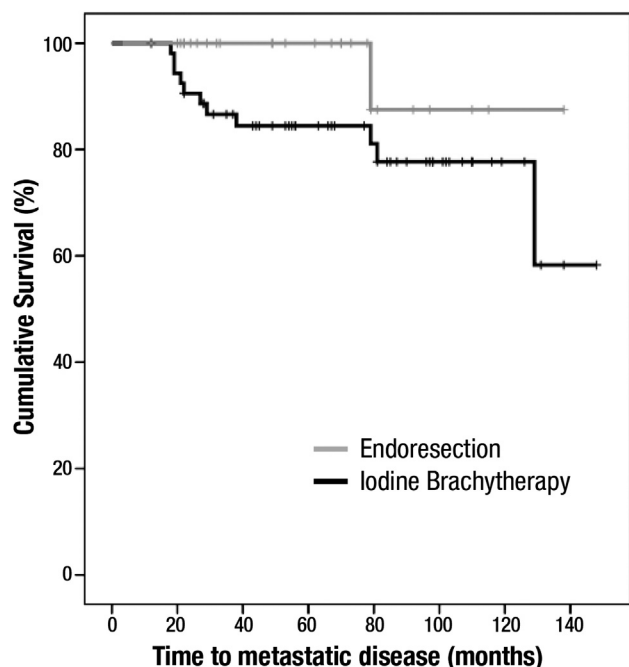


FIGURE 2. Kaplan-Meier analysis of metastasis-free survival after treatment in patients with choroidal melanomas: endoresection vs iodine-125 plaque brachytherapy ($P = .114$, log-rank test).

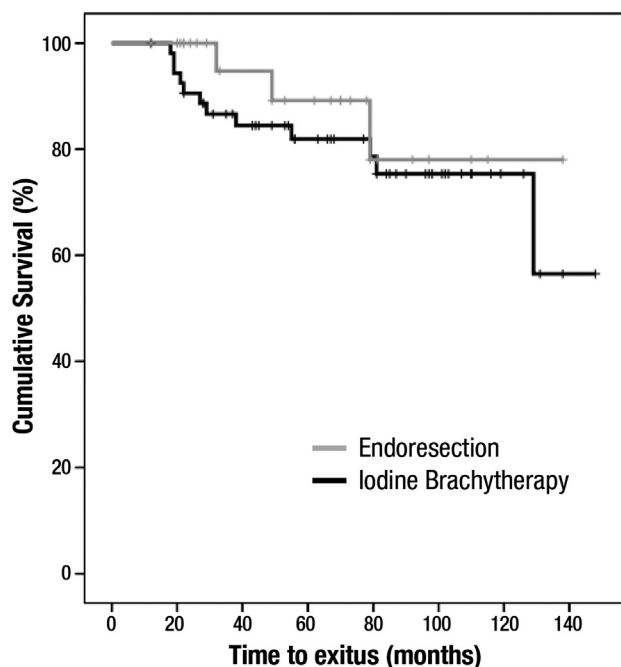


FIGURE 3. Kaplan-Meier analysis of observed survival after treatment in patients with choroidal melanomas: endoresection vs iodine-125 plaque brachytherapy ($P = .429$, log-rank test).

within a median of 119 ± 30 hours (range, 7-197 hours) at a median dose rate of 77 ± 22 cGy/hour (range, 40-141 cGy/hour). The median doses delivered to the optic nerve and fovea were, respectively, 51.9 ± 34 Gy (range, 2-134 Gy) and 64 ± 49 Gy (range, 2-224 Gy).

• **METASTASIS, SURVIVAL, AND LOCAL RECURRENCE:** Metastasis was observed in 11 of the 54 patients in the iodine-125 brachytherapy group and in 1 of 24 patients in the endoresection group (20.4% and 3.7%, respectively; $P = .053$) (Table 3). The estimated Kaplan-Meier proportion of patients without metastatic disease was expected to be 84.2% in the iodine-125 brachytherapy group and 100% in the endoresection group at 5 years follow-up ($P = .114$, log-rank test) (Figure 2).

Fourteen patients died during follow-up; of these, 11 were in the iodine-125 brachytherapy group and 3 in the endoresection group (20.4% and 11.1%, respectively; $P = .238$). Eleven patients died of metastatic disease during follow-up, 10 in the iodine-125 brachytherapy group and 1 in the endoresection group (18.5% and 3.7%, respectively; $P = .089$) (Table 3). Kaplan-Meier analysis of observed survival (all-cause) predicted 81.5% survival in the iodine-125 brachytherapy group and 89.2% survival in the endoresection group at 5 years follow-up ($P = .429$, log-rank test); cause-specific survival was expected to be 84.2% of patients in the iodine-125 brachytherapy group and 100% of patients in the endoresection group at 5 years follow-up ($P = .147$, log-rank test) (Figures 3 and 4).

Three patients experienced local recurrence at the surgical margins during the follow-up period; of these, 1 was in the iodine-125 brachytherapy group and 2 in the endoresection group (1.8% and 7.4%, respectively; $P = .256$) (Table 3). The estimated Kaplan-Meier proportion of patients without local recurrence was 96.6% for the iodine-125 brachytherapy group and 92.4% for the endoresection group at 5 years post treatment ($P = .2$, log-rank test) (Figure 5).

Cox regression analysis for local recurrence was not performed because of the very low number of events. Risk factor analysis for overall survival (HR 0.443, 95% CI 0.083-2.375; $P = .343$) and metastasis-free survival (HR 0.540, 95% CI 0.015-19.631; $P = .736$) revealed that the treatment modality did not affect prognosis regardless of patient age, sex, tumor height, or maximum tumor base. Thus, this multivariate analysis demonstrates that none of these variables was a determinant of patient outcome.

• **VISUAL ACUITY:** Mean visual acuity at diagnosis as measured by Snellen and logMAR units, respectively, was as follows: 0.5 ± 0.7 for the iodine-125 brachytherapy group vs 0.2 ± 0.5 for the endoresection group ($P = .084$) and 0.3 ± 0.9 vs 0.69 ± 2.7 ($P = .084$). Visual acuity was comparable between the 2 groups but was slightly lower in the endoresection group. At the time of diagnosis, best-corrected visual acuity (BCVA) was 20/40 or better in 77.8% and 66.7% of patients, respectively, in the iodine-125 brachytherapy and endoresection groups

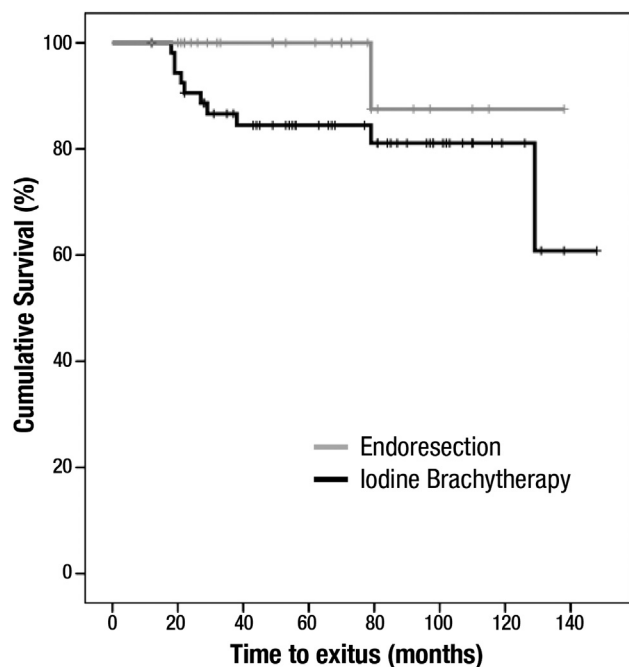


FIGURE 4. Kaplan-Meier analysis of melanoma-specific survival after treatment in patients with choroidal melanomas: endoresection vs iodine-125 plaque brachytherapy ($P = .147$, log-rank test).

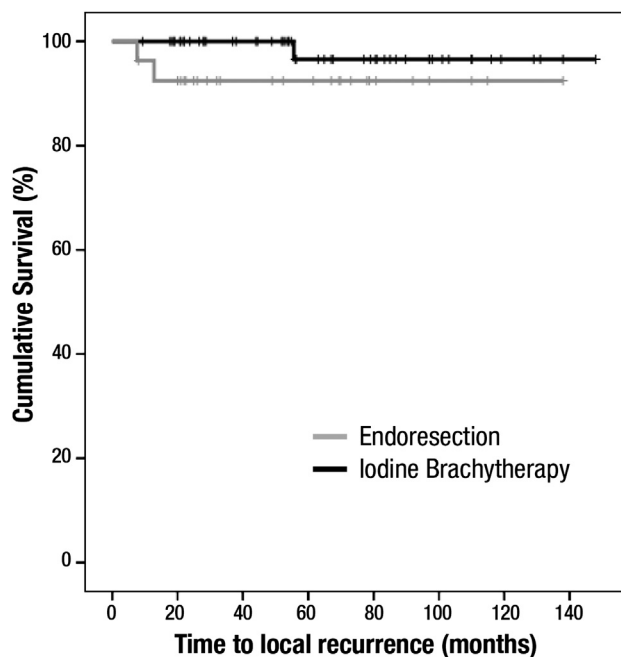


FIGURE 5. Kaplan-Meier analysis of the estimated percentage of choroidal melanoma patients without local recurrence after treatment: endoresection vs iodine-125 plaque brachytherapy ($P = .2$, log-rank test).

($P = .058$) (Tables 1 and 2). Final BCVA was 20/200 or better in 35.2% of patients in the brachytherapy group vs 22.2% in the endoresection group ($P = .234$) (logMAR average of 2.01 ± 1.17 and 2.56 ± 0.83 , respectively; $P = .234$) (Table 3).

According to the Kaplan-Meier estimate, the probability of maintaining a visual acuity equal or superior to 20/200 at 5 years after treatment was 66.4% in the iodine-125 brachytherapy group and 59.9% in the endoresection group ($P = .083$, log-rank test) (Figure 6). Cox regression analysis showed no significant difference in the probability of maintaining a visual acuity equal or superior to 0.1 as a function of treatment (HR 1.393, 95% CI 0.780-2.487), irrespective of age, sex, or distance to the optic nerve or fovea.

• **SECONDARY ENUCLEATION RATES:** Ten patients underwent secondary enucleation: 7 (13%) in the iodine-125 brachytherapy group and 3 (11.1%) in the endoresection group ($P \geq .999$). Three patients underwent secondary enucleation attributable to local tumor recurrence: 1 (1.8%) in the brachytherapy group and 2 (7.4%) in the endoresection group ($P = .183$). Other causes of enucleation were neovascular glaucoma (5 cases; 9.3%) and retinal detachment (1 case; 1.9%) in the iodine-125 brachytherapy group and phthisical eye (1 case; 3.7%) in the endoresection group (Table 3).

According to the Kaplan-Meier estimates, the probability of maintaining the involved eye at 5 years after treatment was 85.7% in the iodine-125 brachytherapy

group and 87.8% in the endoresection group ($P = .942$, log-rank test) (Figure 7). Cox regression analysis showed no significant difference in the enucleation rate between treatments (HR 1.327, 95% CI 0.223-7.902), regardless of patient age, sex, tumor height, and maximum tumor base.

DISCUSSION

THE OPTIMAL THERAPY FOR UVEAL MELANOMA IS STILL a subject of debate, although currently the emphasis is on eye-preserving techniques.^{1,7,11} Comparative studies of the various treatment options are sparse and often problematic because of the rarity of the disease and the heterogeneity of the different tumor characteristics that influence patient survival. We conducted this retrospective study to compare endoresection to iodine-125 brachytherapy, with an emphasis on the most relevant factors (ie, tumor size and location) that affect treatment outcome.¹² Despite the controversy surrounding endoresection, in our study we found no significant differences between groups in terms of metastatic disease, overall and specific survival rates, visual acuity, or secondary enucleation rates.

The main concern about endoresection is that fragmenting the tumor with a vitreous cutter could, theoretically, cause malignant cell seeding, thus potentially prompting systemic tumor dissemination.^{10,13} However, Suesskind and associates¹⁴ carried out a pilot study to assess the

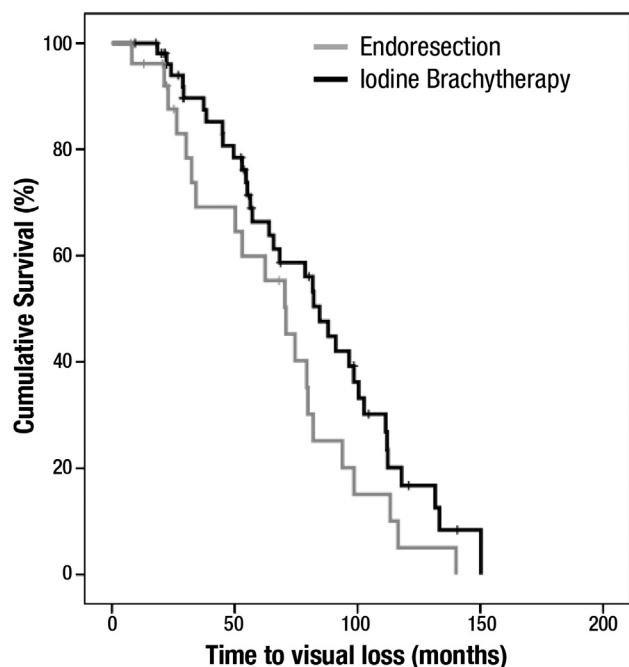


FIGURE 6. Kaplan-Meier analysis of the estimated percentage of choroidal melanoma patients with a visual acuity 20/200 or better after treatment: endoresection vs iodine-125 plaque brachytherapy ($P = .083$, log-rank test).

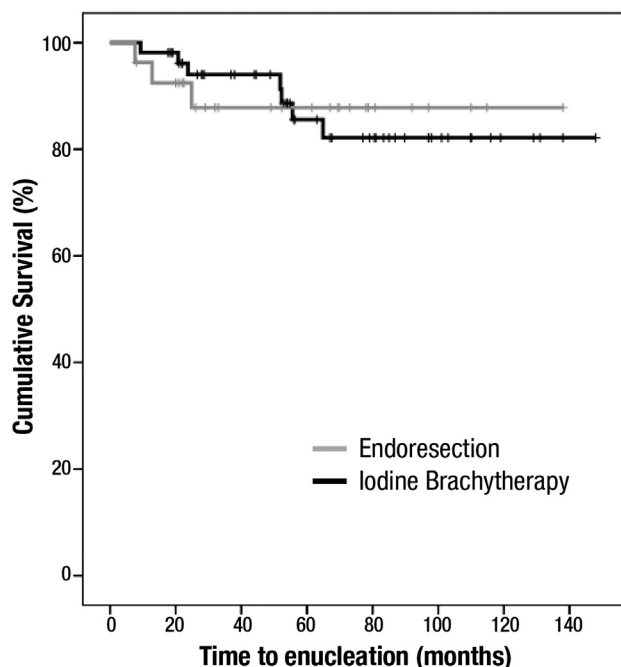


FIGURE 7. Kaplan-Meier analysis of the estimated percentage of choroidal melanoma patients retaining the involved eye after treatment: endoresection vs iodine-125 plaque brachytherapy ($P = .942$, log-rank test).

presence of circulating melanoma cells and found that endoresection does not appear to produce an increase in these cells. Based on our own results and those of other authors, metastatic disease does not appear to be increased in endoresection. In our study, we actually found a trend towards a lower incidence of metastasis in the patients who underwent endoresection (20.4% vs only 3.7%). Moreover, we found no statistically significant differences between the 2 groups in terms of observed survival rate and cause-specific survival rate after 5 years of follow-up. Similarly, Damato and associates⁵ reported only 1 death (2% of sample) from metastatic disease after endoresection in a group of 52 choroidal melanoma patients. García-Arumi and associates⁷ also reported good results with endoresection: a melanoma-specific survival rate of 90.9% in a 38-patient series. In another study, Karkhaneh and associates¹⁵ reported that 1 patient (5%) died of liver metastasis in their series of 20 patients.

Some authors have proposed preoperative treatment to prevent intraoperative dissemination of tumor cells and distant recurrence. Bechrakis and associates^{9,16,17} performed neoadjuvant proton beam irradiation with good results: the rate of local tumor control was 98%, with a 97% eye retention rate. However, at 5 years follow-up, 21% of patients presented metastasis. Schilling and associates⁸ described results of a study in which pretreatment stereotactic gamma-knife radiosurgery was performed prior to endoresection, reporting that 7 of the 46 patients developed liver metastasis, with 6 deaths over

the course of a short follow-up time of just 410 days. The data from these studies suggest that preoperative external radiotherapy does not prevent tumor cell dissemination.

Another common concern about endoresection is the potential for incomplete tumor resection or the presence of residual intrascleral melanoma.^{10,13} In a histopathologic review of eyes that were enucleated after endoresection, Hadden and associates¹⁸ found that tumor recurrence occurred at the site of the primary tumor, except for 1 case of seeding. García-Arumi and associates⁷ found local recurrence requiring enucleation in 2 out of 34 patients (5.8%) before the third year of follow-up. Damato and Wong¹⁹ described a case of intrascleral recurrence of a uveal melanoma following endoresection with laser photocoagulation of the tumor margins. These authors suggested that the recurrence may have been caused by the inherent limitations of laser photocoagulation, which may not have removed all cancerous cells, and they concluded that plaque brachytherapy would be a more effective adjuvant therapy. Our own results were similar to those reported by the studies described above: the incidence of local recurrence was low in both treatment groups, although it was nonsignificantly higher in the endoresection group. We suspect that this higher rate occurred because we applied laser treatment only at the surgical margins and the scleral bed, without the use of adjuvant brachytherapy. This fact has prompted us to change our approach, and we now routinely use adjuvant brachytherapy after endoresection based on the premise that local tumor recurrence in our

patients most likely occurred as a result of tumor spread attributable to incomplete resection.

In posteriorly located uveal melanomas, it is difficult to maintain visual acuity because of proximity to the optic nerve and fovea. A posterior tumor location, especially in close proximity to the optic disc and macula, is a risk factor for severe visual acuity loss in patients treated with either endoresection or iodine-125 brachytherapy.^{20,21} In the study by Garcia-Arumi and associates,⁷ a post-treatment BCVA less than 20/400 was reported in 27 of 38 cases, with only 2 patients maintaining 20/100 or better BCVA. In a study by Damato and associates,⁵ the BCVA of 18 of 36 patients ranged from counting fingers to 20/400. This disparity in visual acuity retention might be explained at least in part by differences in tumor size and proximity to the optic disc and fovea, both factors associated with more pronounced vision loss.²² Unsurprisingly, the final post-treatment BCVA was worse than 20/200 in most of our patients, regardless of the treatment arm. Results in the endoresection group were slightly worse: only 22% retained 20/200 or better visual acuity vs 35% in the iodine-125 brachytherapy group. However, the tumors in the endoresection group were significantly closer to the fovea and optic nerve, and a larger (though not statistically significant) proportion of patients in this group presented with poor visual acuity at diagnosis.

In our study, the secondary enucleation rate was low in both groups (13% for brachytherapy and 11% for endoresection). Consequently, the probability of maintaining the ocular globe at 5 years follow-up was high for patients in both groups (86% and 88%, respectively). Other authors have reported secondary enucleation rates between 5.8%

and 15% for patients treated with endoresection.^{5,7,15} In our study, the primary reasons for enucleation in the brachytherapy group were treatment complications (neovascular glaucoma and massive retinal detachment), while the primary reason for enucleation in the endoresection group was local recurrence.

There are some limitations to this study. Although we considered tumor location and tumor height when selecting patients for inclusion in the study, other prognostic factors may influence the prognosis of these patients. In addition, the small number of patients in this study makes it difficult to draw any definitive conclusions from our final results. Furthermore, even though the mean follow-up time was 5 years in both groups, metastatic disease has been described beyond this period. Finally, this is an observational study that reflects the experience of a single center. An ideal comparison would require a multicenter, prospective randomized controlled study (phase III study), which would pose many technical challenges.

Despite these limitations, endoresection seems to be a reasonable, eye-preserving alternative to traditional approaches. This technique may be especially suited for patients with thicker tumors that are posteriorly located in close proximity to the optic disc or fovea, because such patients have an increased risk of visual loss with radiation therapy and would therefore benefit from an ab interno approach. As surgical techniques evolve and surgeons gain more experience with this technique, it is likely that better final visual acuity results will be achieved. Prospective studies with larger patient samples are required before establishing a definitive role for endoresection in the treatment of posterior uveal melanomas.

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Biosketch

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